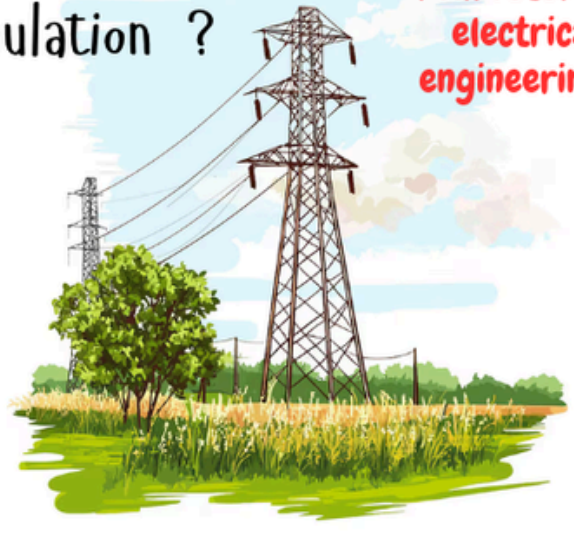
The IDEA of Virtual Synchronous Generator



Can we create an actual genrator with simulation ?

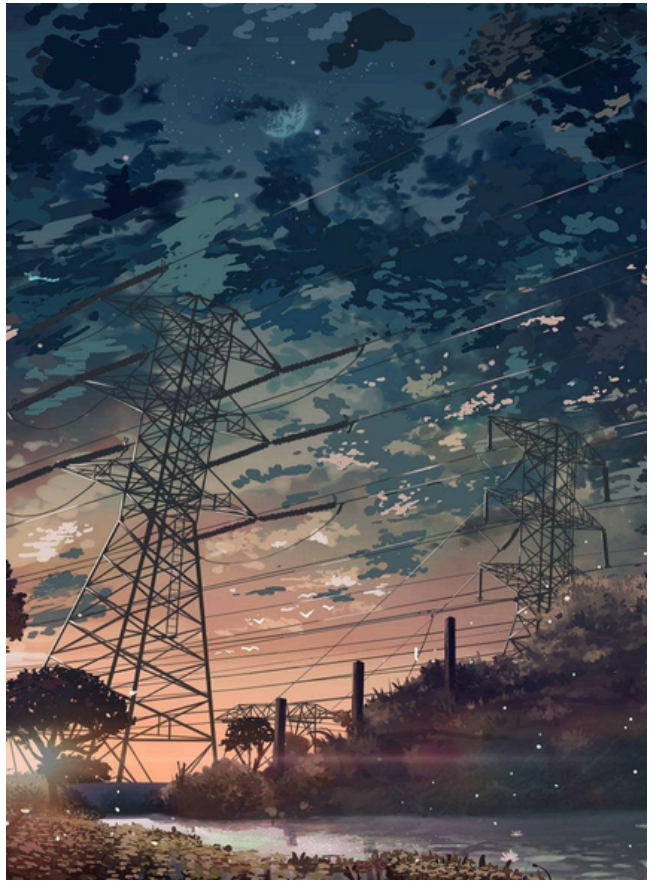
Computer simulation is the real future of electrical engineering ?





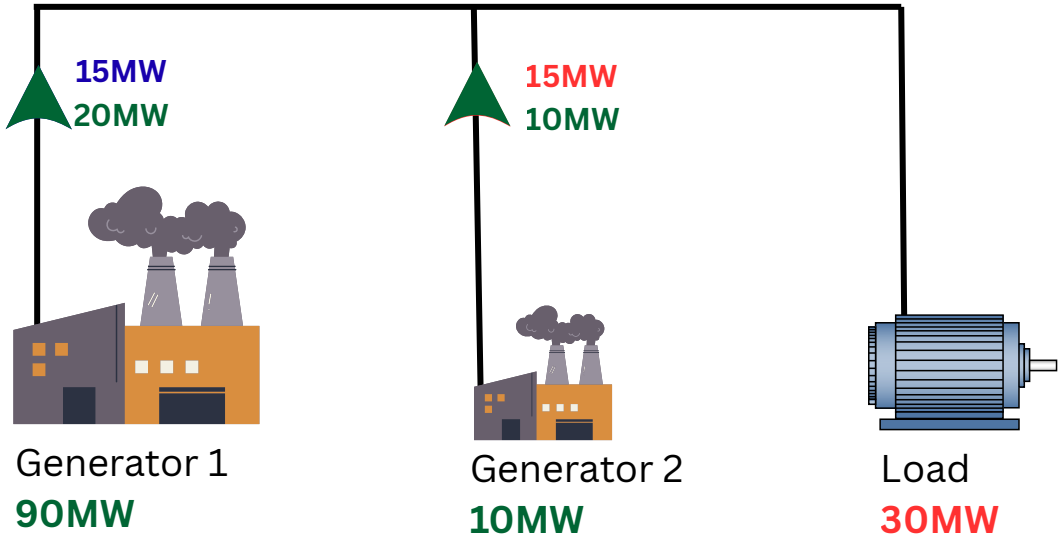
Presentation on

The synchronization of the solar energy with the gird and virtual synchronous generators



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Synchronization of two electrical generators



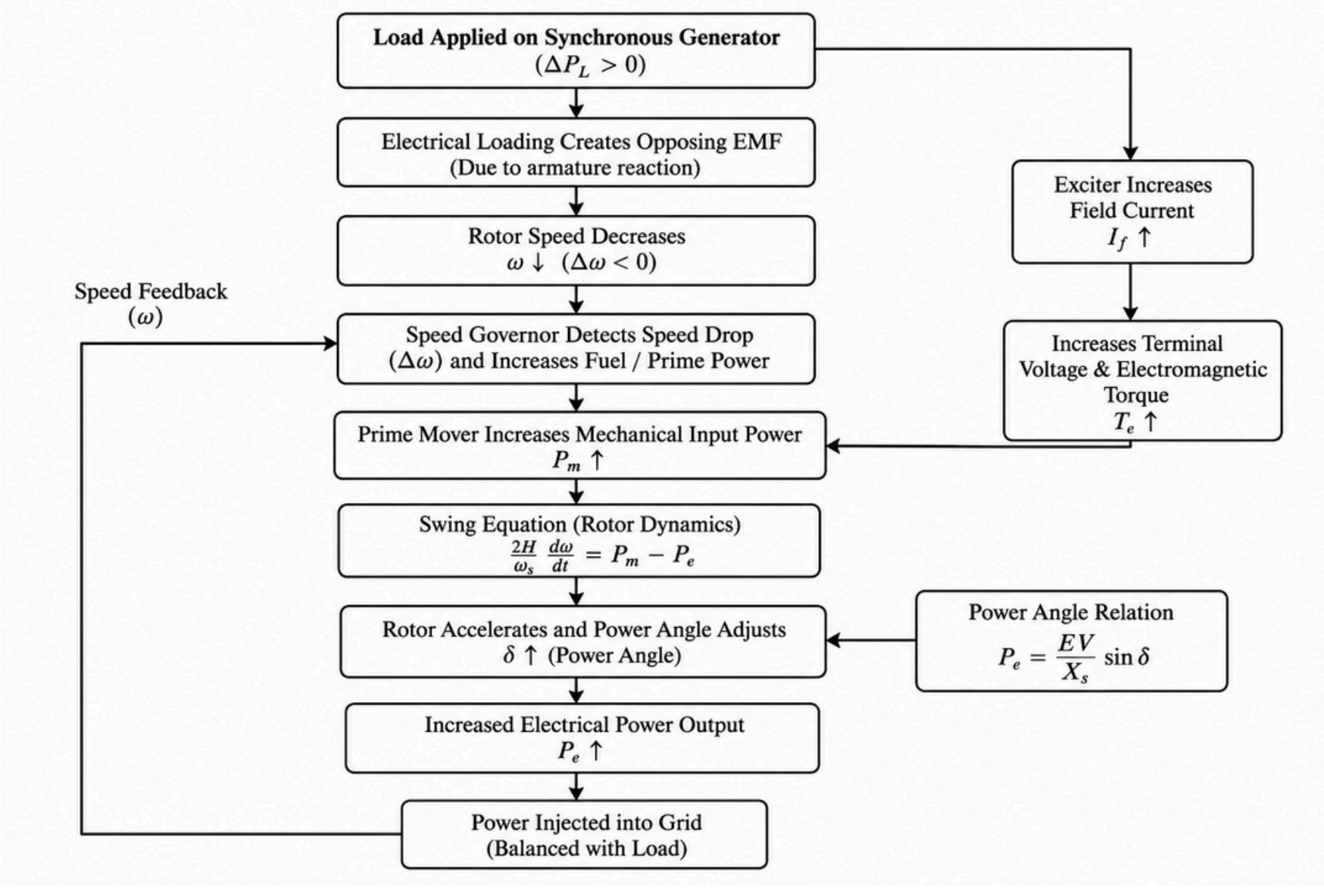


How can we control the power transfer form the generator to the grid ?

If we reduce the power generation of the generator 2 then how would the generator 1 know that ?

To understand the power controlling of the synchronous generator **Swings equation** and **Power Angle Equation.**

Electrical loading and governing of synchronous generator



Power transfer and control of synchronous generators

Swings Equation -

$$J\frac{d\omega_m}{dt} = T_m - T_e$$

This equations give us the relation that how the electrical loading after the rotor speed

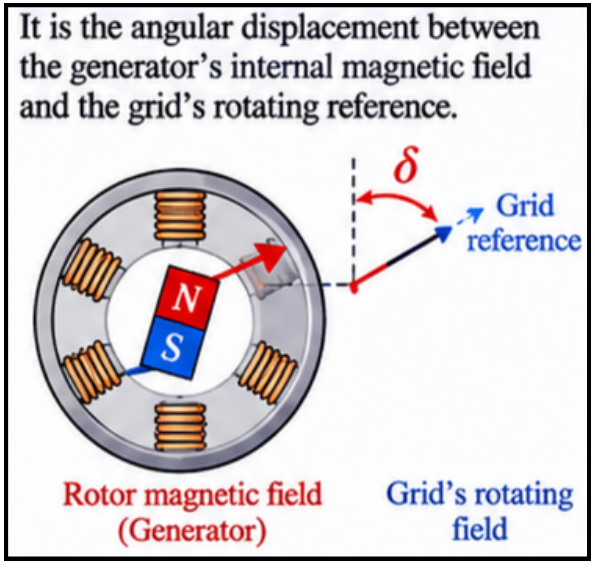
Where:

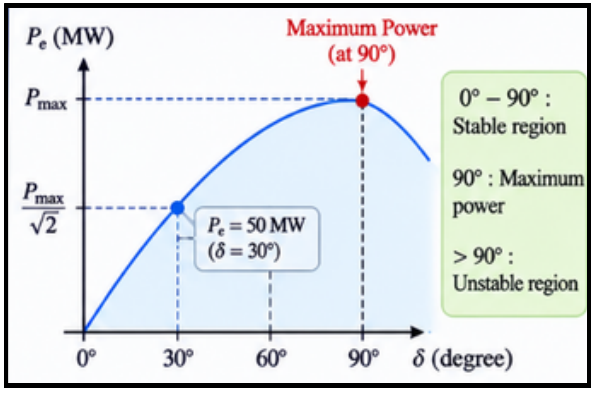
- T_m = Mechanical torque supplied by the prime mover.
- T_e = Electromagnetic torque opposing rotor rotation.
- J = Moment of inertia.
- ω_m = Rotor mechanical angular speed.

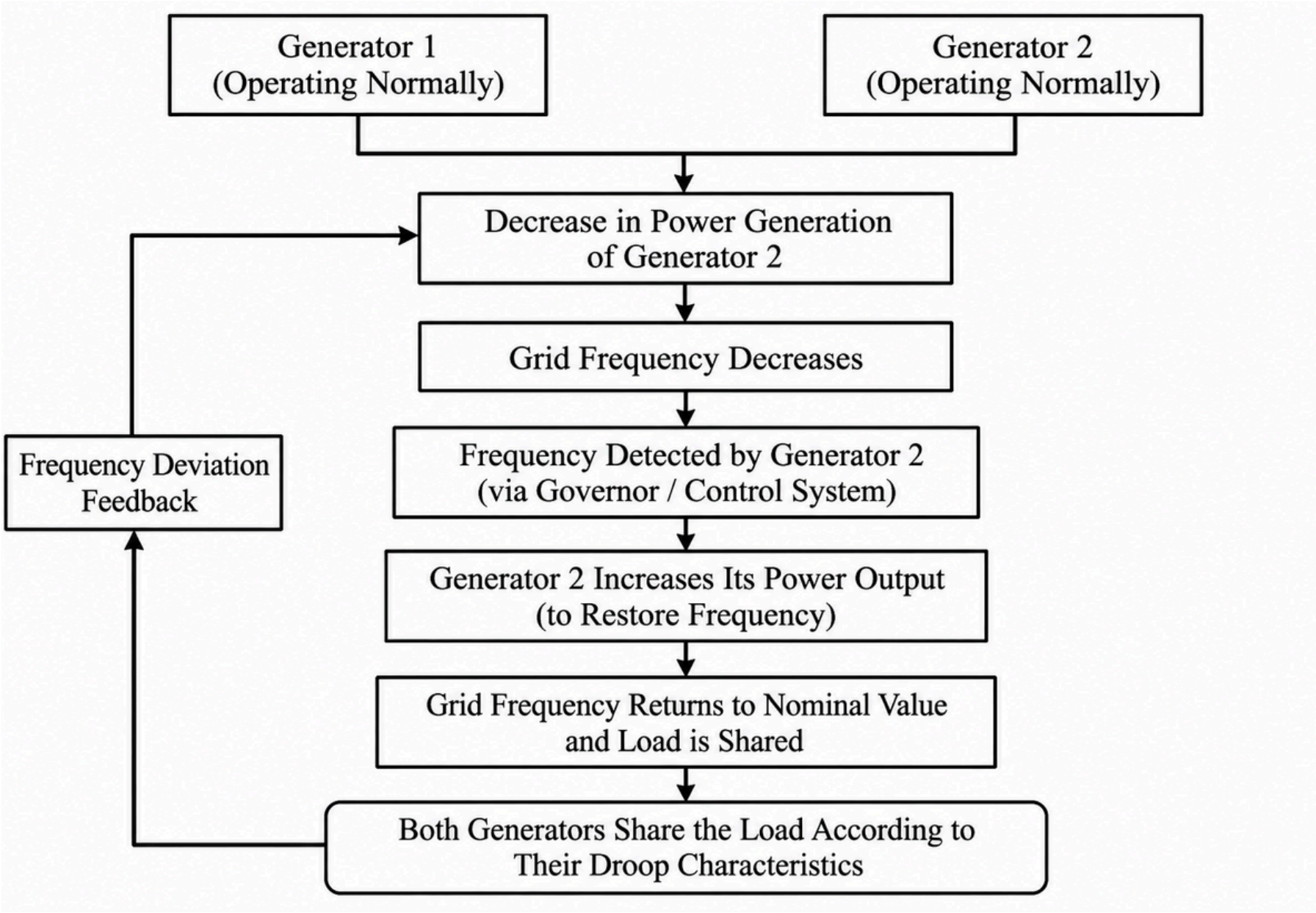
$$P_e = \frac{E_L V_L}{X_s}\sin\delta$$

This equation tell us that how we can control the power transfer of the generator using Power Angle

- E = Internal generated EMF of the generator.
- V = Grid terminal voltage.
- X_s = Synchronous reactance.
- δ = Power angle (angle between E and V).






Generator 1
(Operating Normally)
Generator 2
(Operating Normally)
Decrease in Power Generation
of Generator 2
Grid Frequency Decreases
Frequency Detected by Generator 2
(via Governor / Control System)
Generator 2 Increases Its Power Output
(to Restore Frequency)
Grid Frequency Returns to Nominal Value
and Load is Shared
Both Generators Share the Load According to
Their Droop Characteristics
Frequency Deviation
Feedback



But how would the generators know that how much energy they have to produce ?

or who decide that which generator need to generate how much of electricity ?

To understand it we have to understand the Droop Equations of the synchronous generators ?

Droop Equations / Characteristics -

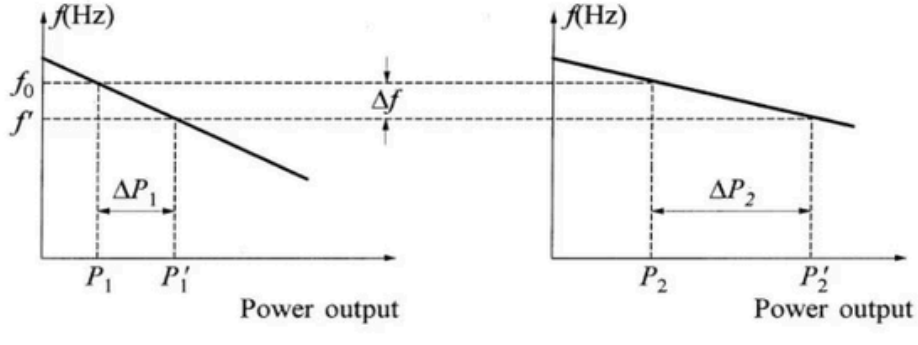
The Droop characteristics provide a way for **multiple generators to automatically share an increasing electrical load** without requiring continuous communication between them.

Load sharing between two synchronous machine in parallel

- Hence

$$\frac{\Delta P_1}{\Delta P_2} = \frac{R_2}{R_1}$$

- If the percentage of regulation of the units are nearly equal, the change in the outputs of each unit will be nearly in proportion to its rating

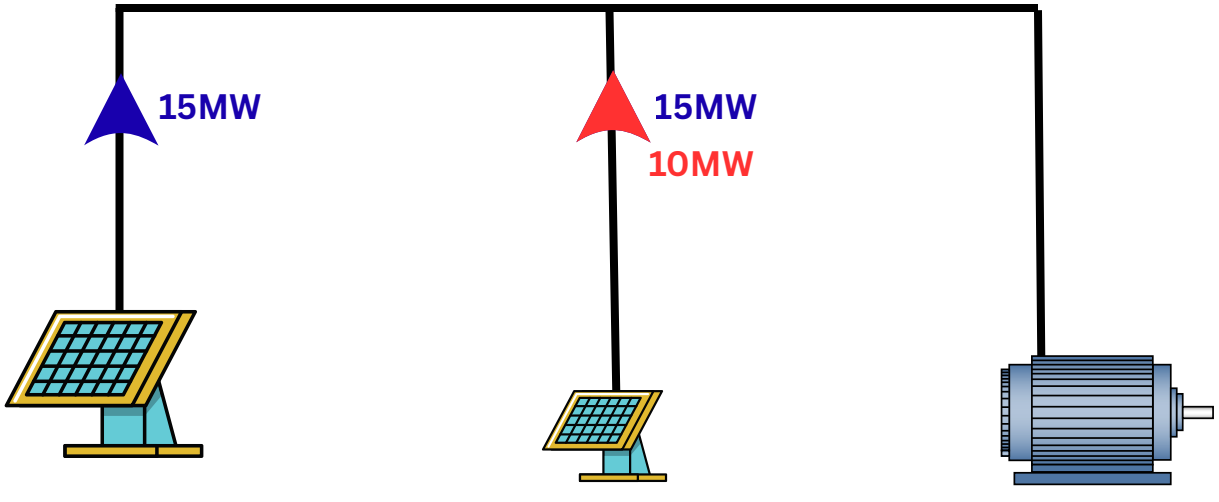


Load sharing by parallel units with drooping characteristics

The droop characteristics of a synchronous generator are primarily determined by the settings and design of its governor control system rather than by its manufacturing date or the connected electrical load. The governor controls the mechanical input supplied to the prime mover (such as steam, water, or fuel) according to changes in system frequency.Engineers configure the governor's droop setting during commissioning

What happen in the case of only solar generating plants

Suppose both the inverter operate at the constant frequency 50hz and synchronized to each other

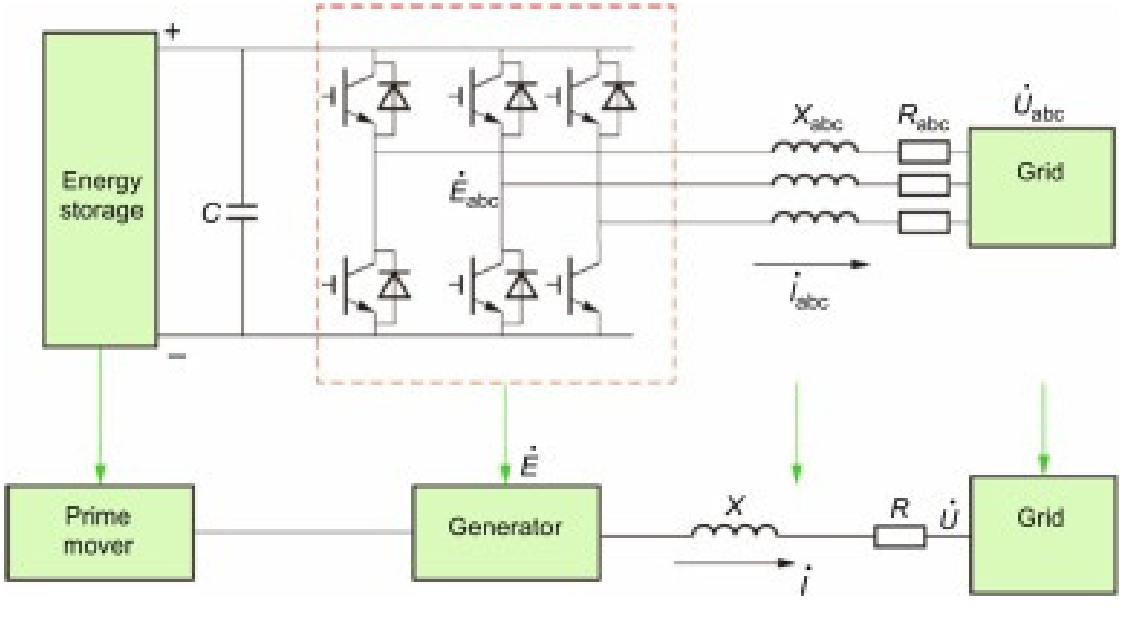


Now here the big question is that the solar inverters do not have any actual rotating rotor it means the frequency not decrease when the load increased on the grid and if the frequency is not decreasing then then is not any droop characteristics exists then how does the each plant or inverter know that how much power they have to generate independently without communicating to each other.

To solve this problem we have an technology name **Virtual Synchronous Generator**

Virtual Synchronous Generator

A Virtual Synchronous Generator (VSG) is a software-based control technique that uses mathematical algorithms to emulate the dynamic behaviour of a conventional synchronous generator, including inertia, damping, and power-angle characteristics. It enables inverters to respond to grid frequency variations like synchronous generators and implement frequency-power droop control, allowing multiple inverters to share electrical loads automatically while improving grid stability.



The virtual generator has the battery storage system to recreate the rotor inertia in the gird